Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** translating business requirements into **structured platform workflows**, **SQL-backed configurations**, and client-facing implementations, with strong alignment to **documentation**, **testing**, **defect resolution**, and design-standard enforcement.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Designed a **multi-tenant data platform** on Azure SQL and Blob Storage for **500+ users**, converting operational requirements into normalized schemas, validation rules, and access-controlled workflows.
- Built **FastAPI** integration pipelines across PostgreSQL, SQL Server, and cloud storage, processing **5K+ monthly jobs** with reconciliation checks, issue triage, and **99%+ production reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested a **Flask + React** grading platform serving **150+ instructors**, translating rubric and submission requirements into configurable dashboard and API workflows for **10K+ submissions**.
- Documented grading flows, validated edge cases across PDF/DOCX/ZIP ingestion, and maintained **GitHub Actions CI/CD** for **10K+ requests/day** while reducing turnaround time by **60%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservice software with **Jenkins**, PyTest, and Mockito, maintaining requirements-linked build, test, and release flows across **3 operating systems** with **20+ unit/integration suites**.
- Worked with Microsoft/GitHub stakeholders to document and refine API and ETL changes using **AWS Glue**, **S3**, and OOP patterns, improving response time by **30%** and pipeline efficiency by **68%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Configured a **MERN** resource-management platform with **REST APIs**, WebSockets, and microservices, supporting **8+ business workflows**, role-based access, and **2M+ synchronized records**.
- Mapped structured entities for users, resources, customers, invoices, and notifications into **Azure SQL** while using Jenkins-driven validation and defect testing to improve release reliability and reduce wastage by **42%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built a cross-platform booking platform with **React**, TypeScript, **Flask**, and Firebase, documenting **12+ backend integration points** and translating client workflow requirements into booking, coupon, and payment configurations.
- Validated sub-**200ms** API flows across **50K+ booking records**, tracked webhook defects, and improved booking confirmation speed by **40%** while reducing manual refund handling by **70%**.

Skills

Languages: Python, Java, C++, JavaScript/TypeScript, SQL, XML
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets, Microservices
Frontend & Design Tools: React.js, Next.js, TailwindCSS, AutoCAD-adjacent design workflows
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, Firebase
Platforms & Tooling: Microsoft Excel, Git, SVN-style version control, GitHub Actions, Jenkins, Docker
Delivery & Validation: Requirements Documentation, SIT/UAT Support, Defect Tracking, Unit & Integration Testing, Technical Documentation, Agile
Data & Workflow Systems: SQL-backed Configuration, Business Rules, ETL Pipelines, Structured Record-Keeping, Client Workshops

Projects

Chat2SQL Live | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting **10+ databases**, combining Java + TypeScript services, schema exploration, and structured query-management workflows from a unified interface.

CPU-scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*
- Implemented **6 scheduling algorithms** with configurable controls and Gantt-based output across **1K+ simulated process combinations**, strengthening rule-based workflow validation and defect analysis.

HMIS Live | *MERN, TypeScript, Twilio, Appwrite*
- Built a healthcare workflow platform supporting **100+ daily appointment workflows**, integrating consent-based access, structured records, and monitored user-facing operations that reduced retrieval time by **2x**.